

TECHNICAL NOTE

Title:	Analysis of Lacq Controlled Release Results
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1. SUMMARY

Flylogix was invited to participate in a blind controlled release experiment at Lacq between 16th - 20th September 2024, organised by Stanford University and TotalEnergies. Flylogix set up on Monday 16th and completed 18 surveys from the 17th to the 19th of September.

Prior to the trial Flylogix noted the following limitations. They were unable to fly their fixed wing beyond visual line of sight aircraft for the trial. This meant they could not achieve with the same separation, speed and grid density as their commercial service. They did not have access to high accuracy wind data generated by their aircraft at every altitude. These limitations were recorded and supplied to Stanford and TotalEnergies in FX-TN-034 prior to the trial. Nonetheless, Flylogix embraced the test opportunity and attempted to replicate the offshore method as closely as possible using a quadcopter operated by a third-party Drone Service Provider flying a more limited flight profile.

This report compares the performance of the Flylogix method against the true methane emission rates for each survey period. Flylogix have also been given wind data from a Zephyr Lidar instrument and reprocessed using this data. The report investigates the correlation, likely discrepancies and how this will translate to Flylogix's standard offshore method.

2. COMPARISON OF REPORTED RESULTS

In Figure 1 below we show the Release Data for the Surveys that Flylogix completed:

Performer	Expt ID	Week	Day	BT = BLIND TEST	Test N°	Test number	Release Start	Release Start stabilized	Release End	Duration (hh:mm)	Gas	Localisation : TADI North (N) TADI South (S)	Leak equipment	Leak Type / Location	Type of emission: Horizontal (H) Vertical (V)	Leak localisation			Flowrate								
							(local time)	(local time)	(local time)								X (m)	Y (m)	Z (m)	(g/s)	(kg/h)						
						Tests STANFORD - IMEO - W38/D2	Morning: Mist ; Temp: 11-12°C ; Wind: 1-2 m/s, SouthWest																				*
1	W38	17/09/2024	BT	1	2024W38/17-01		09:00:30	09:01:00	09:46:00	00:45	CH4	North	Scrubber	Flange			21.5	-4.1	2.3	0.25	0.9						
2	W38	17/09/2024	BT	2	2024W38/17-02		09:59:00	10:00:30	10:46:00	00:45	CH4	North	Large Tank D1	Man hole			10.3	-8.1	3.0	2.03	7.3						
							Afternoon: More and more cloudy ; Temp: 22-23°C ; Wind: 1-2 m/s, SouthWest																				
3	W38	17/09/2024	BT	5	2024W38/17-05		13:00:00	13:01:45	13:46:30	00:44	CH4					#N/A	#N/A	#N/A	0.00	0							
4	W38	17/09/2024	BT	6	2024W38/17-06		13:58:30	14:00:00	14:45:00	00:45	CH4	South	Fixed Point 1	Vertical tubing	V		22.7	-39.7	1.1	15.00	54						
5	W38	17/09/2024	BT	7	2024W38/17-07		14:58:00	14:58:15	15:44:00	00:45	CH4	North	Product Skid	Connector			15.5	-3.4	1.4	0.01	0.02						
						Tests STANFORD - IMEO - W38/D3	Morning: A little bit cloudy ; Temp: 11-12°C ; Wind: 1-2 m/s, EastSouthEast																				
6	W38	18/09/2024	BT	1	2024W38/18-01		08:51:30	08:52:00	09:37:00	00:45	CH4	North	Pipe Rack	Drilled Plug			24.4	-7.5	0.6	0.14	0.5						
7	W38	18/09/2024	BT	2	2024W38/18-02		09:44:30	09:45:30	10:31:00	00:45	CH4	South	Fixed Point 1	Horizontal tubing	H		22.7	-39.7	1.8	10.69	38.47						
8	W38	18/09/2024	BT	4	2024W38/18-04		11:44:00	11:45:00	12:33:00	00:48	CH4	South	Fixed Point 1	Horizontal tubing	H		22.7	-39.7	1.8	14.71	52.94						
							Afternoon: Sunny, some clouds ; Temp: 25-26°C ; Wind: 1-2 m/s, East																				
9	W38	18/09/2024	BT	5	2024W38/18-05		12:45:00	12:45:30	13:25:30	00:40	CH4	North	Small Tank D2	Valve			21.8	-7.4	0.4	0.08	0.29						
10	W38	18/09/2024	BT	7	2024W38/18-07		14:29:00	14:29:30	15:15:45	00:46	CH4	North	Large Tank D1	Valve			10.0	-11.7	0.5	0.05	0.19						
11	W38	18/09/2024	BT	8	2024W38/18-08		15:30:00	15:31:30	16:17:00	00:45	CH4	South	Fixed Point 2	Vertical piping	V		21.7	-39.5	1.3	80.56	290						
12	W38	18/09/2024	BT	9	2024W38/18-09		16:31:20	16:33:00	17:19:45	00:46	CH4	North	Product Skid	Control Box			15.1	-2.4	1.2	0.20	0.73						
						Tests STANFORD - IMEO - W38/D4	Morning: Partly cloudy ; Temp: 14°C ; Wind: 1-2 m/s, East NorthEast																				
13	W38	19/09/2024	BT	1	2024W38/19-01		08:45:00	08:46:00	09:32:30	00:46	CH4	South	Fixed Point 1	Horizontal tubing	H		22.7	-39.7	1.8	7.98	28.71						
14	W38	19/09/2024	BT	2	2024W38/19-02		09:45:00	09:45:30	10:31:00	00:45	CH4	North	Small Tank D2	Man hole			21.5	-8.7	2.4	0.45	1.63						
							Afternoon: Sunny, some clouds ; Temp: 23-25°C ; Wind: 2-3 m/s, East																				
15	W38	19/09/2024	BT	5	2024W38/19-05		12:44:00	12:45:00	13:32:30	00:47	CH4	South	Fixed Point 1	Horizontal tubing	H		22.7	-39.7	1.8	13.61	49.01						
16	W38	19/09/2024	BT	6	2024W38/19-06		13:46:00	13:47:00	14:34:00	00:47	CH4	North	Well Head	South flange			26.1	-14.3	1.5	0.32	1.14						
17	W38	19/09/2024	BT	8	2024W38/19-08		15:44:30	15:46:30	16:34:00	00:47	CH4	North	Flare (6.5 m)	Open Ended Line	V		37.1	-6.9	6.5	2.29	8.25						
18	W38	19/09/2024	BT	9	2024W38/19-09		16:44:10	16:45:00	17:36:00	00:51	CH4	South	Fixed Point 1	Horizontal tubing	H		22.7	-39.7	1.8	7.33	26.4						

Figure 1: Release Data from the Trials that we completed and submitted results

Comparing the Release Data with our Reported Results we see that in Table 1 the Flylogix method reported results sympathetic to the size of the emission with no bias at the zero and low emission levels.

Performer Experiment ID	Estimated Emission Rate Lower [kg/h]	Estimated Emission Rate [kg/h]	Estimated Emission Rate Upper [kg/hr]	True Emission Rate [kg/h]
1	0.0	0.3	2.2	0.90
2	0.0	1.7	7.5	7.30
3	0.0	0.0	0.3	0.00
4	0.0	10.3	23.9	54.00
5	0.0	0.0	0.4	0.02
6	0.0	0.3	2.3	0.50
7	0.0	17.7	42.8	38.47
8	16.4	38.6	60.8	52.94
9	0.0	0.0	0.9	0.29
10	0.0	0.1	0.9	0.19
11	142.7	171.9	201.1	290.00
12	0.0	0.1	1.0	0.73
13	0.0	12.4	26	28.71
14	0.0	0.7	5.4	1.63
15	0.0	11.3	25.4	49.01
16	0.0	1.1	3.4	1.14
17	0.0	2.9	5.5	8.25
18	2.6	5.6	8.6	26.40

Table 1: Comparison between Reported Results and the True Emission Rate

The rows in red {4, 11, 13, 15, 17 and 18} show where the True Emission value is outside of the Reported Result Error bounds. There is a significant discrepancy on the highlighted results.

The graphical representations are shown in Figure 2 and zoomed in in Figure 3 and show significant discrepancy on only 4 highlighted results {4, 11, 15 and 18} below:

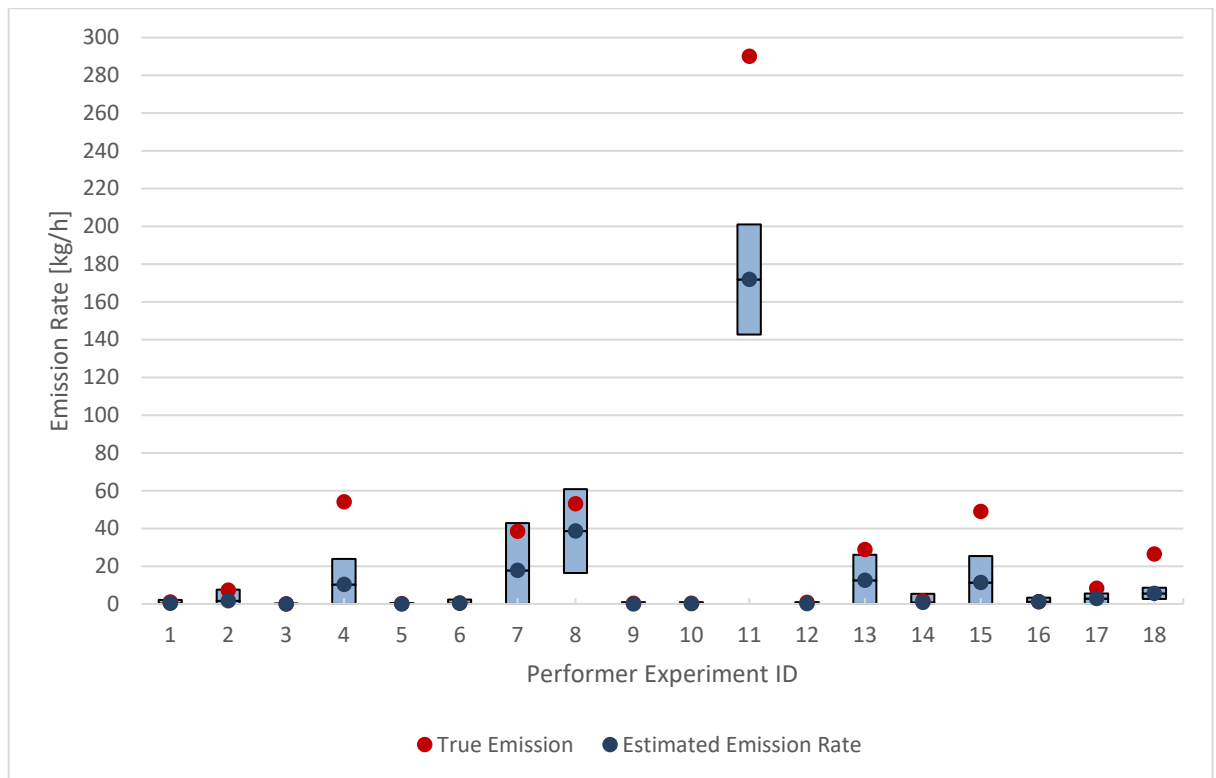


Figure 2: Comparison of Reported Results with the True Emission rates

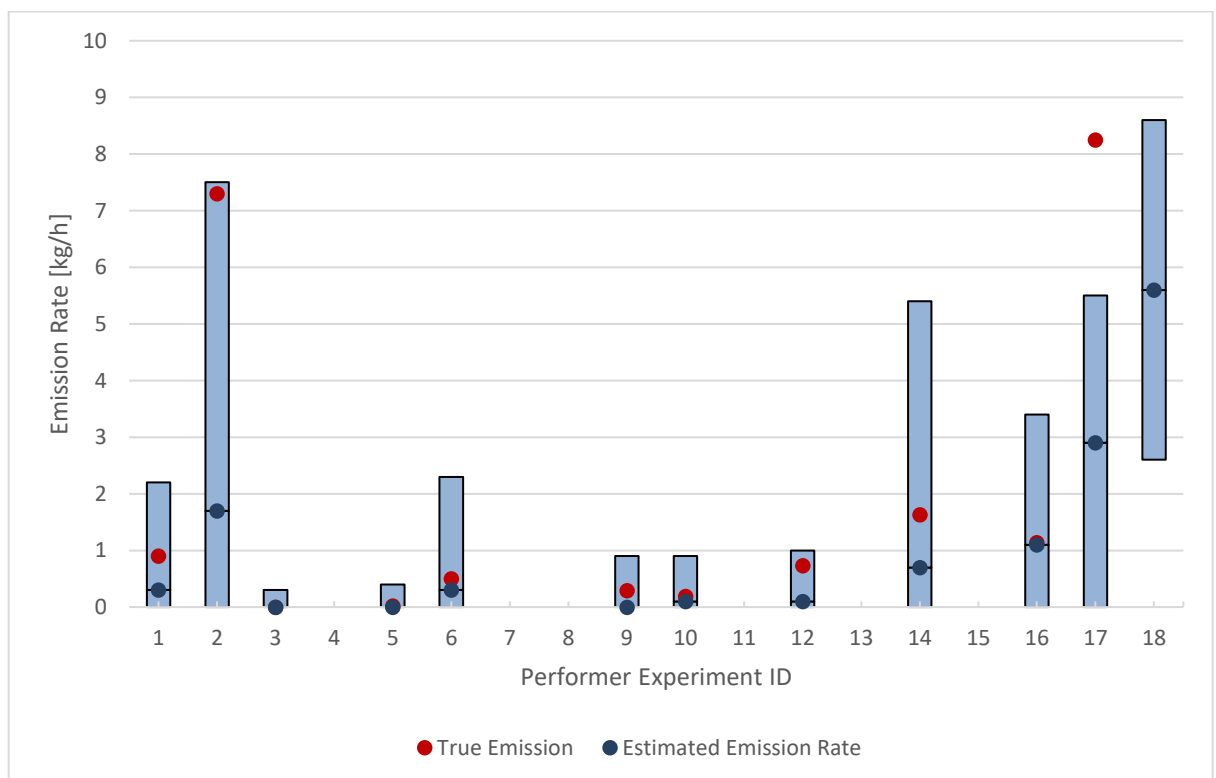


Figure 3: Zoomed in comparison of Reported Results vs True

3. COMPARISON OF RESULTS USING LIDAR WIND DATA

The results were reprocessing using the Zephyr Lidar wind data at 20m using estimated 3σ wind speed errors and wind direction errors of ± 1 m/s and $\pm 10^\circ$ respectively in order to mitigate the poor-quality wind data Flylogix were forced to use for this trial. The results in Table 2 show an improvement.

Performer Experiment ID	Estimated Emission Rate Lower [kg/h]	Estimated Emission Rate [kg/h]	Estimated Emission Rate Upper [kg/hr]	True Emission Rate [kg/h]
1	0.0	0.4	1.8	0.90
2	0.0	3.0	6.6	7.30
3	0.0	0.0	0.2	0.00
4	7.5	14.8	22.1	54.00
5	0.0	0.0	0.3	0.02
6	0.0	0.5	2.1	0.50
7	17.0	34.0	51.0	38.47
8	36.1	52.7	69.3	52.94
9	0.0	0.0	0.4	0.29
10	0.0	0.2	0.8	0.19
11	238.5	259.8	281.1	290.00
12	0.0	0.1	0.8	0.73
13	15.2	25.5	35.8	28.71
14	0.0	0.9	3.9	1.63
15	8.6	20.6	32.6	49.01
16	0.0	1.3	3.0	1.14
17	1.8	4.9	8.0	8.25
18	5.1	7.6	10.1	26.40

Table 2: Comparison between Results using LIDAR wind data and the True Emission Rate

However, there are still 6 results in red {2, 4, 11, 15, 17 and 18} that fall outside of our error bounds.

The graphical representations are shown in Figure 4 and Figure 5 and show a significant discrepancy on only 3 highlighted results {4, 15 and 18} below:

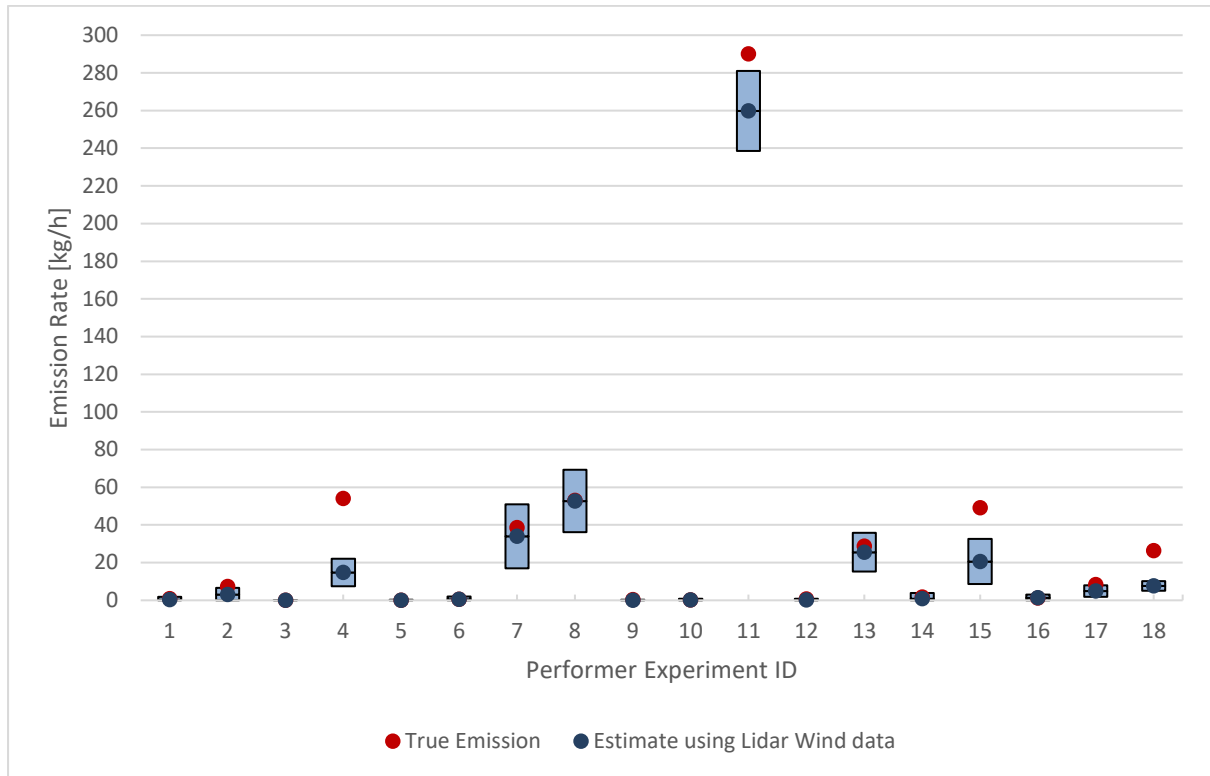


Figure 4: Comparison of reprocessed results using the Lidar wind data

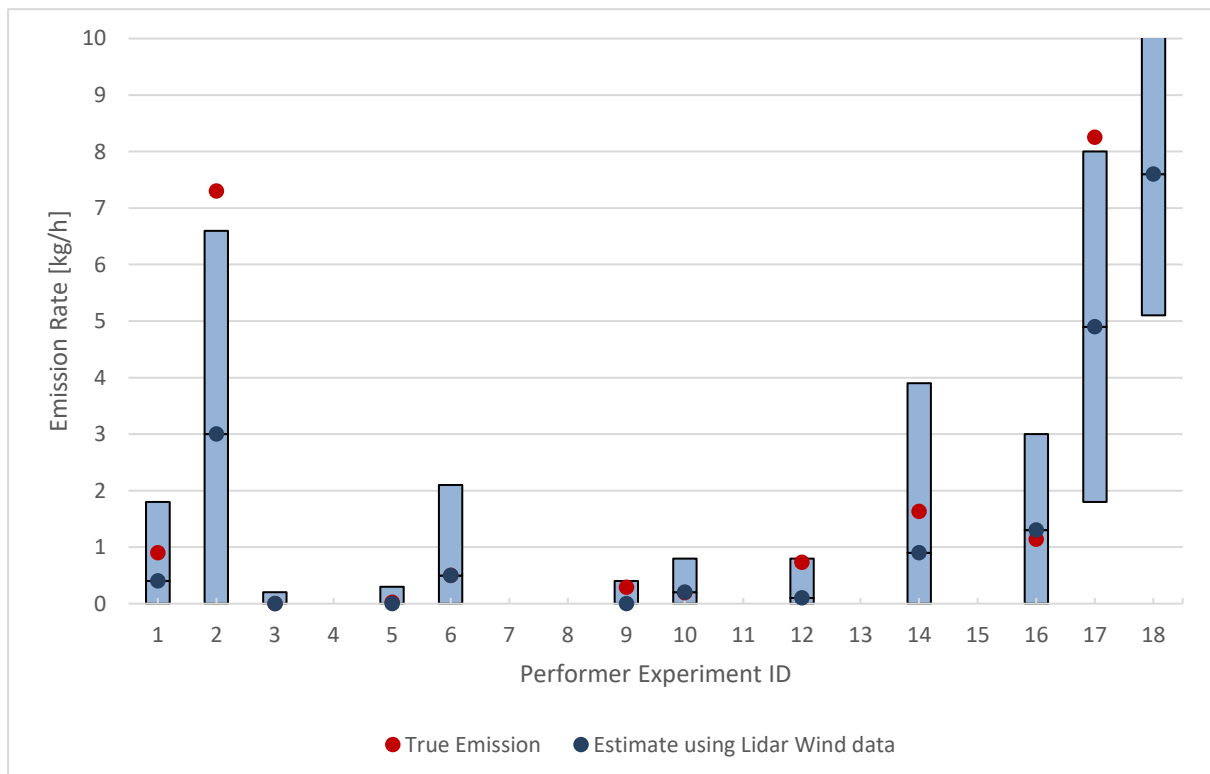


Figure 5: Zoomed in comparison of results using the Lidar wind data

4. DISCUSSION

The results show that we generally underestimated the true emission rates using the Davis Weather Station at 2m above the ground level. The results improved when using the Lidar Wind data at 20m: the bias is significantly reduced indicating that the cup anemometer was unable to accurately measure wind speed and direction in these low wind speed conditions and was located too close to ground level.

However, the results for rows highlighted {4, 15 and 18} did not improve with the Lidar wind data results. Sections 4.1, 4.2 and 4.3 investigate these significant discrepancies in more detail:

4.1 Experiment ID #4

Figure 6 shows the emissions from Experiment ID #4. We see that there are still methane detections at the top and bottom of the survey, indicating that there might have been methane escaping above and below the flight path due to the very low wind conditions. This suggests we simply missed part of the emission at the top and bottom of the survey and need to extend the upper and lower altitude of the survey in low wind conditions. Flylogix were able to extend the upper altitude of later surveys but were restricted from coming lower by the site.

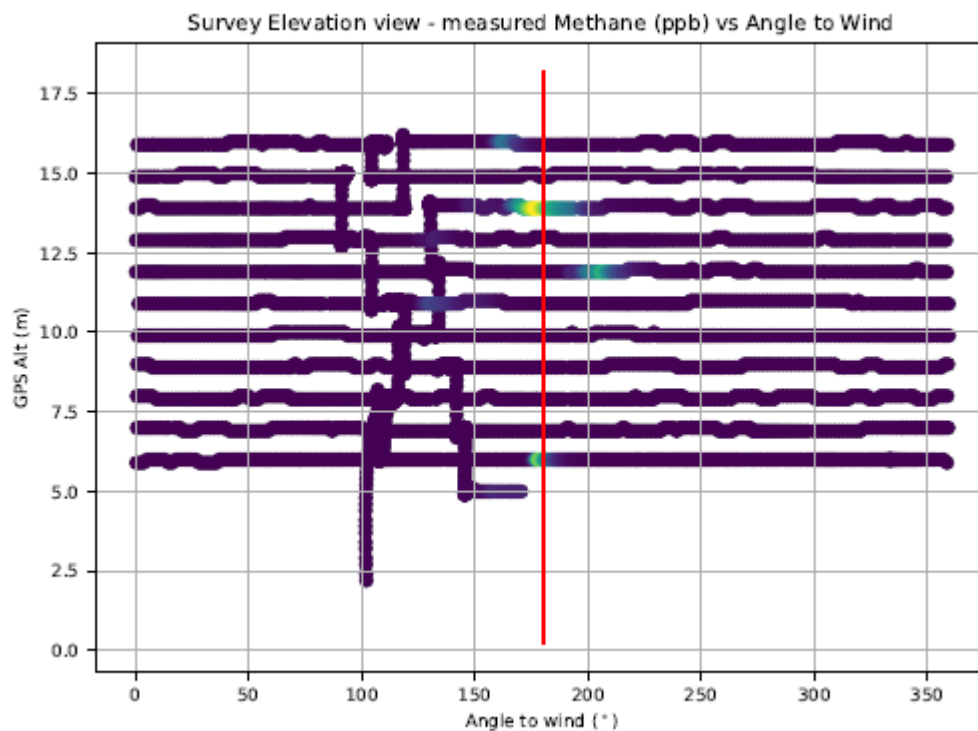


Figure 6: Experiment ID #4 – potentially missing methane out top and bottom

4.2 Experiment ID #15

Figure 7 shows the flight path from Experiment ID #15. We see that UAV was changing altitude exactly downwind of the release as circled in green in Figure 7:

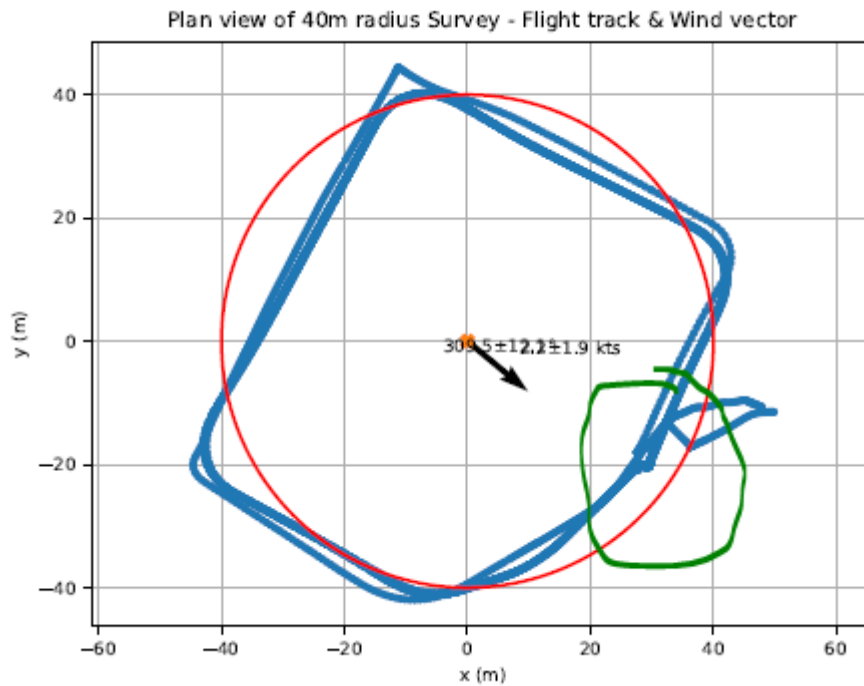


Figure 7: Experiment ID #15 – altitude changes occur downwind of the release

Because the UAV's speed drops to zero during the altitude change, the data is discarded to stop 'double counting' of methane. This means we removed critical measured emissions and reported a lower emission. This is unique to limitations of the quadcopter operation and is not an issue in a Flylogix survey.

4.3 Experiment ID #18

Figure 6 shows the emissions from Experiment ID #18. We see that we have one very large detection at a low altitude. In this case we seem likely to have missed some methane at the lower altitude and need to fly lower. Unfortunately, it was not possible to fly lower due to site restrictions.

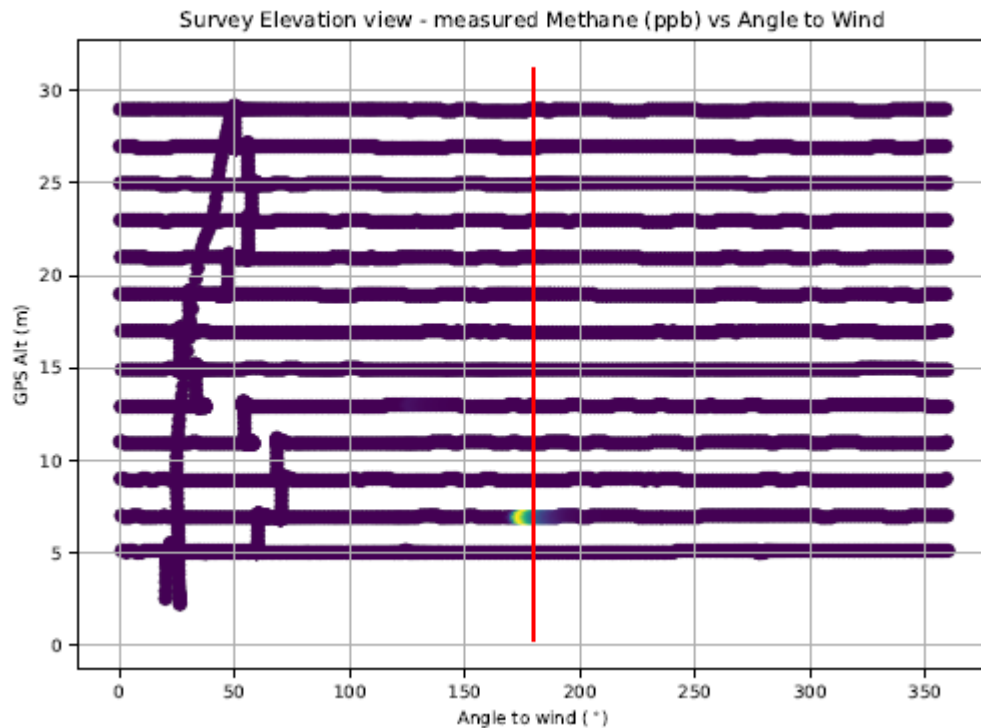


Figure 8: Experiment ID #18 – potentially missing methane at low altitude

5. CONCLUSION

Flylogix were not able to use their standard fixed wing beyond visual line of sight aircraft flying a large diameter cylinder to conduct this trial but still accurately measured an emission in 12 out of 18 tests in challenging low and variable wind conditions using a quadcopter flying a simplified cylinder around the emitter. Flylogix were able to improve this to 15 out of 18 tests by using the LIDAR wind data provided by Stanford to mitigate the lack of aircraft wind data.

For the three measurements where a significant discrepancy remained, Flylogix believe that part of the emission passed above or below the measurement cylinder in two cases and that the flight path of the drone led to loss of emissions data in the third case. These limitations are not present in Flylogix's standard offshore measurement method. Therefore Flylogix believe they would have achieved 18 accurate measurements if they had been able to use their standard method.

Flylogix also believe the higher accuracy wind data from the aircraft combined with the greater grid density due to greater aircraft speed and longer measurement duration creates greater accuracy in their offshore work. In addition, Flylogix are able to fly a much taller cylinder, reducing the risk of emissions escaping. Finally, the greater speed allows Flylogix to stand back further to allow the plume to develop and become more stable, further improving the accuracy of the measurement.

Flylogix appreciate the opportunity to test and refine their understanding of the method and welcome an opportunity to conduct a representative test of their offshore method.